

OSTROVNOE, Russian Federation

WMO: 251380

Lat: 68.112N

Lon: 164.173E

Elev: 94

StdP: 100.20

Time Zone: 12.00 (E12)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-46.6	-45.0	-49.5	0.0	-45.8	-48.1	0.0	-45.0	9.8	-3.2	8.5	-12.4	0.5	80	0.786

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	26.3	15.7	24.2	15.0	22.2	14.3	17.3	23.9	16.0	21.9	14.8	20.7	2.4	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.6	10.5	19.7	13.2	9.6	19.0	11.9	8.7	17.8	49.2	24.2	45.4	21.9	41.9	20.9	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.3	6.4	5.5	-48.3	30.2	1.5	2.3	-49.4	31.8	-50.3	33.1	-51.1	34.4	-52.2	36.0		
			-48.3	19.5	1.5	2.1	-49.4	21.0	-50.3	22.2	-51.1	23.4	-52.2	24.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-9.1	-32.2	-31.7	-21.8	-9.5	2.9	12.6	13.9	10.4	3.8	-8.4	-20.7	-29.5
	DBStd	18.43	10.08	8.43	9.16	7.40	5.84	4.47	4.73	4.12	4.36	6.66	9.00	9.63
	HDD10.0	7271	1309	1167	986	586	229	27	17	45	193	569	921	1224
	HDD18.3	10017	1568	1400	1244	836	479	176	151	248	436	828	1171	1482
	CDD10.0	314	0	0	0	0	8	104	138	57	7	0	0	0
	CDD18.3	19	0	0	0	0	0	3	14	2	0	0	0	0
	CDH23.3	272	0	0	0	0	0	66	181	25	1	0	0	0
	CDH26.7	55	0	0	0	0	0	3	51	1	0	0	0	0
	WSAvg	2.2	1.8	1.7	2.2	2.3	2.7	2.6	2.5	2.3	2.3	2.2	2.0	1.8
Precipitation	PrecAvg	238	11	11	9	6	13	31	38	37	27	20	19	15
	PrecMax	297	23	31	28	13	29	63	71	77	74	45	37	33
	PrecMin	170	2	1	0	0	1	9	6	9	5	4	2	4
	PrecStd	33	6	7	6	3	8	12	17	18	16	11	9	8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.7	-0.1	3.2	7.4	21.1	27.5	30.7	26.0	20.7	7.5	1.1	0.4
		MCWB	-0.2	-2.0	0.7	2.7	11.8	14.3	19.4	16.5	12.3	4.3	-0.9	-1.3
	2%	DB	-1.9	-6.3	-1.2	5.4	17.6	25.5	28.0	23.6	16.7	4.5	-0.6	-4.6
		MCWB	-3.9	-7.5	-3.6	1.3	10.1	13.9	17.7	15.2	10.7	2.9	-2.2	-5.8
	5%	DB	-11.4	-16.0	-5.3	3.7	14.6	23.6	25.1	21.1	13.4	2.1	-4.1	-10.4
		MCWB	-12.0	-16.5	-6.9	0.0	8.3	13.7	16.3	14.2	8.9	0.8	-5.1	-11.1
	10%	DB	-17.5	-21.1	-8.9	1.8	12.2	21.6	22.7	18.7	10.7	-0.1	-7.2	-15.6
		MCWB	-18.0	-21.5	-10.4	-1.2	6.5	13.1	15.3	12.8	7.7	-0.9	-7.9	-16.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.1	-2.0	0.7	3.1	12.8	16.7	20.9	17.7	13.2	4.6	-0.6	-1.1
		MCDB	2.6	-0.1	3.1	7.1	19.4	24.0	29.5	24.1	18.8	6.8	0.7	0.2
	2%	WB	-3.9	-7.8	-3.5	1.4	10.1	15.2	18.3	16.2	11.0	2.9	-2.3	-5.8
		MCDB	-1.9	-6.8	-1.3	5.1	16.9	22.1	25.5	21.9	15.5	4.4	-0.7	-4.6
	5%	WB	-12.0	-16.9	-6.8	0.2	8.5	14.3	17.0	14.8	9.5	0.9	-5.1	-11.2
		MCDB	-11.5	-16.5	-5.1	3.5	14.6	21.8	23.5	20.2	12.8	2.0	-4.1	-10.4
	10%	WB	-18.0	-21.5	-10.4	-1.2	6.7	13.4	15.8	13.3	7.7	-0.9	-7.9	-16.0
		MCDB	-17.5	-21.2	-9.0	1.6	11.9	20.7	21.3	17.6	10.4	-0.1	-7.2	-15.6
Mean Daily Temperature Range	5% DB	MDBR	7.0	7.5	11.0	12.5	9.7	10.7	10.2	10.6	8.2	6.1	6.8	6.7
		MCDBR	13.0	11.2	9.3	10.5	13.4	14.7	14.9	15.4	14.2	6.4	7.6	11.3
		MCWBR	11.8	10.0	8.0	7.5	7.7	6.1	7.1	8.0	8.4	4.7	6.6	10.7
	5% WB	MCDBR	13.1	10.3	9.3	9.9	12.7	12.8	13.6	13.8	12.4	6.1	7.7	11.3
		MCWBR	11.9	9.7	8.0	7.2	7.5	6.1	7.5	7.5	7.7	4.4	6.8	10.6
Clear-Sky Solar Irradiance	taub	0.165	0.222	0.294	0.344	0.380	0.365	0.416	0.398	0.320	0.257	0.183	N/A	N/A
	taud	1.923	2.086	2.061	1.975	2.127	2.387	2.238	2.245	2.435	2.161	1.921	N/A	N/A
	Ebn at Noon	131	591	719	779	800	835	764	731	721	520	86	0	0
	Edh at Noon	8	46	88	130	126	101	114	101	64	42	7	0	0
All-Sky Solar Radiation	RadAvg	0.03	0.55	2.27	4.67	5.78	5.69	4.92	3.63	1.92	0.69	0.10	0.00	0.00
	RadStd	0.00	0.06	0.11	0.14	0.53	0.39	0.46	0.38	0.17	0.07	0.01	0.00	0.00

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	+1.74	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page